

1. Purpose

This document explains the full clinicalguide-ai implementation and exactly what to replicate or inject into n8n so the chatbot behaves like a doctor-facing clinical pathway assistant.

2. Existing clinicalguide-ai architecture

Stack:

- Frontend: Next.js + TypeScript
- Backend logic: API routes + server libraries
- LLM providers: Gemini or Ollama (configurable)
- Embeddings: Gemini embedding model or Ollama nomic-embed-text
- Vector store: Chroma (in clinicalguide-ai), target in MedAssistant: Qdrant
- Document parsing: PDF text extraction and chunking
- Chat memory: local DB (SQLite in clinicalguide-ai)
- Audit trail: stored user query, retrieved chunk ids, response payload

Core files reviewed:

- lib/server/rag.ts
- lib/server/gemini.ts
- app/api/chat/route.ts
- lib/server/db.ts
- lib/types.ts

3. Intelligence patterns already implemented in clinicalguide-ai

3.1 Retrieval-first behavior

- Every question is embedded.
- Top-K chunks are retrieved.
- Similarity threshold and lexical overlap filtering reduce off-topic matches.

3.2 Controlled response schema

The model is forced into JSON:

- assistant_reply
- next_actions (max 1)
- clarification_questions (max 1)
- insufficient_evidence (boolean)

3.3 Safety and quality checks

- Reject echo/repeat responses.
- Reject ultra-short/low-value responses.
- Fallback when evidence is insufficient.

3.4 Conversation memory

- Session-based message history is loaded and reused for follow-up questions.

3.5 Auditability

- Persist retrieved chunk ids and final response payload for traceability.

4. Gaps to close for pathway-level clinical reasoning

Current system is strong RAG chat, but pathway selection is implicit.

To achieve doctor-assistant behavior for management pathways, add explicit reasoning stages:

- Condition identification from query and history

- Branch selection criteria extraction (severity, red flags, special populations)
- Pathway branch selection
- Management step generation linked to chosen pathway

5. Target n8n solution design

Use two workflows:

Workflow A: Guideline Ingestion (PDF -> chunks -> vectors)

- 1) Trigger: Local File Trigger on /data/shared/rag-files/pending/*.pdf
- 2) Read binary PDF
- 3) Parse via Docling HTTP endpoint
- 4) Chunk text with overlap
- 5) Create embeddings using Ollama (nomic-embed-text)
- 6) Upsert vectors to Qdrant collection clinical_guidelines
- 7) Save metadata: source file, chunk index, section/pathway tags if detected
- 8) Move processed files to /data/shared/rag-files/processed

Workflow B: Doctor Assistant Chat (RAG + pathway reasoning)

- 1) Trigger: n8n Chat Trigger
- 2) Build session context from recent conversation store
- 3) Condition extraction node (structured JSON)
- 4) Embed query
- 5) Qdrant similarity retrieval
- 6) Evidence filter node (threshold + coverage)
- 7) Pathway selection node (structured JSON)
- 8) Final response generation node (strict output schema)
- 9) Audit logging node (MySQL)
- 10) Chat response node

6. Structured outputs to enforce in n8n

Condition extraction schema:

- suspected_conditions: string[]
- presentation_summary: string
- red_flags: string[]
- missing_critical_data: string[]

Pathway selection schema:

- selected_condition: string
- candidate_pathways: string[]
- selected_pathway: string
- selection_rationale: string
- evidence_chunk_ids: number[]
- confidence: number
- requires_clarification: boolean
- clarification_question: string

Final assistant response schema:

- assistant_reply: string
- next_actions: string[] (max 1)
- clarification_questions: string[] (max 1)
- insufficient_evidence: boolean
- citations: {chunk_id, source_name, chunk_index}[]

7. Recommended prompt policy for n8n model node

System policy:

- You are a clinical decision support assistant for clinicians.
- Use only retrieved guideline evidence.
- If evidence is insufficient, return conservative response and one focused clarification question.
- Identify probable condition from presentation.
- If multiple pathways exist, explicitly choose one branch and state criteria.
- Return concise clinical guidance and red flags.
- Never fabricate doses/protocols absent from evidence.
- Include citations per statement where possible.

8. MySQL schema for n8n reference and audits

Table: guidelines

- id (PK)
- name (unique)
- version
- uploaded_at

Table: guideline_chunks

- id (PK)
- guideline_id (FK)
- chunk_index
- content
- embedding_json
- section_title
- pathway_tag
- created_at

Table: pathway_nodes

- id (PK)
- guideline_id (FK)
- condition_name
- pathway_name
- node_type (entry, decision, action, red_flag, escalation)
- node_text
- created_at

Table: pathway_edges

- id (PK)
- from_node_id (FK)
- to_node_id (FK)
- selection_rule
- created_at

Table: chat_sessions

- id (PK)
- session_id (unique)
- clinician_id
- created_at

Table: chat_messages

- id (PK)
- session_id

- role (user/assistant)
- text
- created_at

Table: chat_audits

- id (PK)
- session_id
- user_query
- inferred_condition
- selected_pathway
- retrieved_chunk_ids_json
- response_payload_json
- created_at

9. Retrieval and chunking recommendations

- Chunk size: 1000-1400 chars
- Overlap: 180-250 chars
- Top-K: 4-6
- Minimum similarity: tune per model, start around 0.20
- Add keyword overlap guard for long clinical terms

10. Follow-up question handling

For each turn:

- Load last 8-12 messages from session
- Preserve unresolved clarifications
- Ask at most one focused follow-up question
- When enough criteria are present, choose pathway and provide next management step

11. Minimum viable n8n credential setup

- Ollama credential: base URL <http://ollama:11434>
- Qdrant credential: URL <http://qdrant:6333>
- MySQL credential: local DB for audits and sessions

12. Validation checklist before go-live

- Upload a guideline PDF and confirm chunk count > 0
- Confirm vectors inserted in Qdrant
- Test direct condition question
- Test ambiguous presentation requiring clarifying question
- Test pathway branch selection (e.g., stable vs unstable)
- Verify citations and audit logs are written
- Verify insufficient evidence behavior is conservative

13. Example doctor queries for acceptance testing

- "Adult with chest pain and hypotension, what is immediate management pathway?"
- "Child with fever and neck stiffness, next guideline-based steps?"
- "DKA suspected, which branch applies if glucose high and ketones positive?"
- "Postpartum bleeding with tachycardia, first-line pathway?"

14. Implementation priority order in n8n

Priority 1: ingestion + vector indexing

Priority 2: RAG chat + citations

Priority 3: pathway selection stage

Priority 4: MySQL audit + session memory

Priority 5: confidence thresholds and fail-safe responses

15. Final notes

This design keeps the assistant clinically grounded, auditable, and conversational. It enables natural language queries while still forcing structured pathway reasoning under the hood.